Functions_file_handling_By_ChinmayeeNayak

File handling

Q1. Write a Python program to read a file line by line and store it into a list

In [2]:
```
# Solution
# Open the file in read mode
with open("C:\\Users\\nchin\\Desktop\\poem.txt", 'r') as file:
    # Read the file line by line and store each line in a list
    lines = file.readlines()

# Now, the 'lines' variable contains the content of the file as a list, with each l
# we can access and manipulate the lines as needed

# Example: Print the lines
for line in lines:
    print(line, end='')  # Use 'end' to avoid adding extra newlines

# Close the file (it's automatically closed when using 'with' context manager)
```

```
Deep into that darkness peering,

 Long I stood there, wondering, fearing,

 Doubting, dreaming dreams no mortals

 Ever dared to dream before;

 But the silence was unbroken,

 And the stillness gave no token,

 And the only word there spoken

 Was the whispered word, "Lenore!"

 This I whispered, and an echo

 Murmured back the word, "Lenore!"

 Merely this, and nothing more.
```

Q2. Write a Python program to extract characters from various text files and puts them into a list. words1.txt Welcome to the Python Wiki, a user-editable compendium of knowledge based around the Python programming language. Some pages are protected against casual editing - see WikiEditingGuidelines for more information about editing content. words2.txt Python is a great object-oriented, interpreted, and interactive programming language. It is often compared (favorably of course :-)) to Lisp, Tcl, Perl, Ruby, C#, Visual Basic, Visual Fox Pro, Scheme or Java... and it's much more fun.

In [5]:
```
# Solution

# Create a list to store the characters
character_list = []
```

```
# List of file paths
file_paths = ["C:\\Users\\nchin\\Desktop\\words1.txt", "C:\\Users\\nchin\\Desktop\\

# Loop through the files and extract characters
for file_path in file_paths:
    try:
        with open(file_path, 'r') as file:
            content = file.read()
            characters = list(content)
            character_list.extend(characters)
    except FileNotFoundError:
        print(f"File not found: {file_path}")

# Now, character_list contains all the characters from the specified text files
# we can access and manipulate the characters as needed

# Example: Print the first 100 characters
print("First 100 characters:")
print(''.join(character_list[:100]))

# Close the files (they are automatically closed when using 'with' context manager)
```

```
First 100 characters:
Welcome to the Python Wiki, a user-editable compendium of knowledge based around the
Python programm
```

Functions

Q1. Write a Python function to find the Max of three numbers.

In [6]:
```
# Solution

def find_max_of_three(num1, num2, num3):
    max_num = max(num1, num2, num3)
    return max_num

# Example usage:
result = find_max_of_three(10, 5, 15)
print("The maximum number is:", result)
```

```
The maximum number is: 15
```

Q2. Write a Python function to sum all the numbers in a list

In [7]:
```
#Solution
```

```python
def sum_list_numbers(numbers):
    total = sum(numbers)
    return total

# Example usage:
numbers = [1, 2, 3, 4, 5]
result = sum_list_numbers(numbers)
print("The sum of the numbers in the list is:", result)
```

```
The sum of the numbers in the list is: 15
```

Q3. Write a Python function to check whether a string is a pangram or not.

Note : Pangrams are words or sentences containing every letter of the alphabet at least once.

For example : "The quick brown fox jumps over the lazy dog"

In [8]:

```python
# Solution

import string

def is_pangram(input_string):
    alphabet = set(string.ascii_lowercase)
    input_string = input_string.lower()

    for char in input_string:
        if char in alphabet:
            alphabet.remove(char)

    return len(alphabet) == 0

# Example usage:
text = "The quick brown fox jumps over the lazy dog"
result = is_pangram(text)
if result:
    print("The given string is a pangram.")
else:
    print("The given string is not a pangram.")
```

```
The given string is a pangram.
```

OR

In [9]:
```python
def is_pangram(input_string):
    alphabet = set("abcdefghijklmnopqrstuvwxyz")
    input_string = input_string.lower()

    for char in input_string:
        if char in alphabet:
            alphabet.remove(char)

    return len(alphabet) == 0

# Example usage:
text = "The quick brown fox jumps over the lazy dog"
result = is_pangram(text)
if result:
    print("The given string is a pangram.")
else:
    print("The given string is not a pangram.")
```

```
The given string is a pangram.
```

Q4. Write a Python program to sort a list of tuples using Lambda. Original list of tuples: [('English', 88), ('Science', 90), ('Maths', 97), ('Social sciences', 82)] Sorting the List of Tuples: [('Social sciences', 82), ('English', 88), ('Science', 90), ('Maths', 97)]

In [10]:
```python
# Solution

# Original list of tuples
original_list = [('English', 88), ('Science', 90), ('Maths', 97), ('Social sciences

# Sorting the list of tuples using a lambda function based on the second element (t
sorted_list = sorted(original_list, key=lambda x: x[1])

# Print the sorted list
print("Sorting the List of Tuples:")
print(sorted_list)
```

```
Sorting the List of Tuples:
[('Social sciences', 82), ('English', 88), ('Science', 90), ('Maths', 97)]
```

Q5. Write a Python program to get the Fibonacci series between 0 to 50.

In [11]:
```python
# Solution

# Function to generate the Fibonacci series
def generate_fibonacci_series(n):
    fibonacci_series = [0, 1]
    while True:
        next_number = fibonacci_series[-1] + fibonacci_series[-2]
        if next_number > n:
            break
        fibonacci_series.append(next_number)
    return fibonacci_series

# Call the function to generate the Fibonacci series between 0 and 50
fib_series = generate_fibonacci_series(50)
```

```
# Print the generated Fibonacci series
print("Fibonacci series between 0 and 50:")
print(fib_series)
```

```
Fibonacci series between 0 and 50:
[0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
```

OR

In [12]:
```
# Initialize the first two Fibonacci numbers
a, b = 0, 1

# Create an empty list to store the Fibonacci series
fib_series = []

# Generate the Fibonacci series up to 50
while a <= 50:
    fib_series.append(a)
    a, b = b, a + b

# Print the generated Fibonacci series
print("Fibonacci series between 0 and 50:")
print(fib_series)
```

```
Fibonacci series between 0 and 50:
[0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
```

Q6. Write a Python program to convert temperatures to and from celsius, fahrenheit

In [13]:
```
# Solution

# Function to convert Celsius to Fahrenheit
def celsius_to_fahrenheit(celsius):
    fahrenheit = (celsius * 9/5) + 32
    return fahrenheit

# Function to convert Fahrenheit to Celsius
def fahrenheit_to_celsius(fahrenheit):
    celsius = (fahrenheit - 32) * 5/9
    return celsius

# Main program
print("Temperature Conversion:")
print("1. Celsius to Fahrenheit")
print("2. Fahrenheit to Celsius")

choice = input("Enter your choice (1/2): ")
```

```python
if choice == '1':
    celsius = float(input("Enter temperature in Celsius: "))
    fahrenheit = celsius_to_fahrenheit(celsius)
    print(f"{celsius}°C is equal to {fahrenheit}°F")
elif choice == '2':
    fahrenheit = float(input("Enter temperature in Fahrenheit: "))
    celsius = fahrenheit_to_celsius(fahrenheit)
    print(f"{fahrenheit}°F is equal to {celsius}°C")
else:
    print("Invalid choice. Please enter 1 or 2.")
```

```
Temperature Conversion:
1. Celsius to Fahrenheit
2. Fahrenheit to Celsius
Enter your choice (1/2): 1
Enter temperature in Celsius: 30
30.0°C is equal to 86.0°F
```

OR

In [14]:
```python
# Function to convert Celsius to Fahrenheit
def celsius_to_fahrenheit(celsius):
    return (celsius * 9/5) + 32

# Main program
celsius = float(input("Enter temperature in Celsius: "))
fahrenheit = celsius_to_fahrenheit(celsius)
print(f"{celsius}°C is equal to {fahrenheit}°F")
```

```
Enter temperature in Celsius: 35
35.0°C is equal to 95.0°F
```

Q7. Write a Python program to count the number of even and odd numbers from a series of numbers.

In [15]:
```python
# Solution

# Input a list of numbers
numbers = [int(x) for x in input("Enter a list of numbers separated by spaces: ").s

# Initialize counters for even and odd numbers
even_count = 0
odd_count = 0

# Iterate through the numbers and count even and odd numbers
for number in numbers:
    if number % 2 == 0:
        even_count += 1
    else:
        odd_count += 1
```

```
# print the counts
print("Number of even numbers:", even_count)
print("Number of odd numbers:", odd_count)
```

```
Enter a list of numbers separated by spaces: 12 13 14 15 11 10 14
Number of even numbers: 4
Number of odd numbers: 3
```

OR

In [19]:
```
# Predefined list of numbers
numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10,11,12,15,10,18,25,1,3,2,56,88,60]

# Initialize counters for even and odd numbers
even_count = 0
odd_count = 0

# Iterate through the list and count even and odd numbers
for number in numbers:
    if number % 2 == 0:
        even_count += 1
    else:
        odd_count += 1

# Display the counts
print("Number of even numbers:", even_count)
print("Number of odd numbers:", odd_count)
```

```
Number of even numbers: 12
Number of odd numbers: 10
```

Q8. Write a Python program which iterates the integers from 1 to 50. For multiples of three print "Fizz" instead of the number and for the multiples of five print "Buzz". For numbers which are multiples of both three and five print "FizzBuzz".

In [20]:
```
# Solution

for number in range(1, 51):
    if number % 3 == 0 and number % 5 == 0:
        print("FizzBuzz")
    elif number % 3 == 0:
        print("Fizz")
    elif number % 5 == 0:
        print("Buzz")
```

```
    else:
        print(number)
```

```
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
16
17
Fizz
19
Buzz
Fizz
22
23
Fizz
Buzz
26
Fizz
28
29
FizzBuzz
31
32
Fizz
34
Buzz
Fizz
37
38
Fizz
Buzz
41
Fizz
43
44
FizzBuzz
46
47
Fizz
49
Buzz
```

OR

In [21]:
```
for number in range(1, 51):
    output = ""
    if number % 3 == 0:
        output += "Fizz"
    if number % 5 == 0:
        output += "Buzz"

    print(output if output else number)
```

```
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
16
17
Fizz
19
Buzz
Fizz
22
23
Fizz
Buzz
26
Fizz
28
29
FizzBuzz
31
32
Fizz
34
Buzz
Fizz
37
38
Fizz
Buzz
41
Fizz
43
44
FizzBuzz
46
47
Fizz
49
Buzz
```

In [ ]: